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Tuberculosis and Parasitic Infections of the Genitourinary Tract

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Keywords

genitourinary tract; parasitic infection; tuberculosis

Questions

Tuberculosis

1. All of the following mycobacteria cause tuberculosis (TB) EXCEPT:
    a. *Mycobacterium bovis.*
    b. *Mycobacterium avium-intracellulare.*
    c. *Mycobacterium africanum.*
    d. *Mycobacterium microti.*
    e. Bacille Calmette-Guérin (BCG).
2. Which of the following statements about the epidemiology of tuberculosis is FALSE?
    a. TB incidence and mortality have been decreasing worldwide since the year 2000.
    b. Tuberculosis incidence is higher in those who are born outside the United States than in those born in the United States.

c. Prevalence of multidrug-resistant (MDR) tuberculosis cases is now approaching 12% in the United States.
d. The lifetime risk of reactivation of TB is 5% to 10% in most people.
e. Worldwide, tuberculosis is the cause of death in 25% of persons who test positive for human immunodeficiency virus (HIV).

3. Which of the following routes of infection is the most common in genitourinary tuberculosis?
    a. Hematogenous seeding
    b. Lymphatic spread
    c. Direct inoculation
    d. Sexual transmission
    e. Ascending or retrograde infection
4. Which of the following is a not a late complication of genitourinary tuberculosis?
    a. Infertility
    b. Scrotal fistula
    c. Autonephrectomy
    d. Thimble bladder
    e. Papulonecrotic tuberculid
5. Which of the following persons is LEAST likely to have tuberculosis infection?
    a. A patient with fibrosis on chest radiograph and a tuberculin skin test (TST) of 5 mm
    b. A patient with HIV infection and a TST of 3 mm
    c. A recent immigrant from Vietnam with a TST of 11 mm
    d. A BCG-vaccinated patient with a TST of 14 mm
    e. A healthy U.S.-born teacher with a TST of 11 mm
6. Which of the following results most specifically diagnoses genitourinary tuberculosis?
    a. A positive interferon γ release assay
    b. A positive urine polymerase chain reaction (PCR) for *Mycobacterium tuberculosis* complex
    c. A TST reaction of 25 mm
    d. A positive urine acid-fast bacilli (AFB) culture

e. A renal biopsy showing AFB

7. Which of the following first-line antituberculosis agents does not cause hepatic toxicity?
    a. Isoniazid
    b. Rifampin
    c. Pyrazinamide
    d. Ethambutol
    e. Streptomycin
8. Which of the following drugs might have efficacy against extensively drug-resistant (XDR) tuberculosis?
    a. Isoniazid (INH)
    b. Rifampin
    c. Pyrazinamide
    d. Moxifloxacin
    e. Amikacin
9. Which of the urological interventions is emergently indicated?
    a. Nephrectomy of nonfunctional kidney in medically resistant hypertension
    b. Bladder augmentation of a contracted bladder in a patient with severe dysuria
    c. Percutaneous nephrostomy of obstructive hydronephrosis in acute renal failure
    d. Balloon dilatation and ureteral stenting of a proximal ureteral stricture
    e. Boari flap for a lower ureteral stricture that requires excision
10. Which of the following statements is FALSE about genitourinary (GU) TB patients?
    a. Magnetic resonance imaging (MRI) is often used to help diagnose patients with GU TB.
    b. Computed tomography (CT) is most useful in extensive TB disease when other organ systems might be involved.
    c. The most common finding of GU TB on plain film is calcification.
    d. Intravenous urography (IVU) is the best test to detect early renal changes due to TB.

e. The most common finding on IVU is obstructive uropathy from scarring.

Schistosomiasis

1. Of the following drugs, the most effective to treat schistosomiasis is:
    a. albendazole.
    b. praziquantel.
    c. mebendazole.
    d. diethylcarbamazine.
    e. ivermectin.
2. The life cycle stage of *Schistosoma haematobium* that infects humans transdermally is:
    a. the worm.
    b. the schistosomule.
    c. the cercariae.
    d. the egg.
    e. the sporocyst.
3. *S. haematobium* infections are estimated to affect the following number of people:
    a. 1.1 billion
    b. 1.1 million
    c. 900,000
    d. 112 million
    e. 11 million
4. The life cycle stage of *S. haematobium* that induces the majority of human tissue pathology is:
    a. the worm.
    b. the schistosomule.
    c. the cercaria.
    d. the egg.
    e. the sporocyst.
5. The eponym for acute schistosomiasis is:
    a. Katayama fever.

b. Bilharz syndrome.
c. Barlow fever.
d. Toshiro syndrome.
e. Tan's triad.

6. The diagnostic, first-line gold standard for urogenital schistosomiasis is:
   a. polymerase chain reaction.
   b. serology.
   c. cystourethroscopy with bladder biopsy.
   d. rectal biopsy.
   e. urine egg counts.
7. Without treatment, *Schistosoma* worms can live in human hosts for an average of:
   a. 3 months.
   b. 9 months.
   c. 3 to 5 years.
   d. 5 decades.
   e. 5 weeks.
8. Surgical options for reconstruction of irreversible ureteral lesions caused by urogenital schistosomiasis include all of the following EXCEPT:
   a. renal autotransplantation.
   b. Boari flaps.
   c. ureteroureterostomies.
   d. ileal ureter.
   e. suprapubic intravesical ureterostomy.
9. Intermediate snail hosts for *S. haematobium* are members of the following genus:
   a. *Biomphalaria.*
   b. *Oncomelania.*
   c. *Bulinus.*
   d. *Helix.*
   e. *Achatina.*

Other Parasitic Infections

1. The rickettsia-like organism that is an endosymbiont of the parasites that cause lymphatic filariasis (LF) and onchocerciasis is:
    a. *Rickettsia rickettsii.*
    b. *Wuchereria bancrofti.*
    c. *Wolbachia* spp.
    d. *Brugia malayi.*
    e. *Brugia timori.*
2. The chronic manifestations of LF are mostly seen in:
    a. short-term missionaries to endemic areas.
    b. short-term aid workers to endemic areas.
    c. long-term (current) residents of endemic areas.
    d. short-term tourists to endemic areas.
    e. short-term visiting friends and relatives travelers (VFRs) to endemic areas.
3. Most patients infected with *Onchocerca volvulus* live in:
    a. Latin America.
    b. Oceania.
    c. Asia.
    d. Sub-Saharan Africa.
    e. the Middle East.
4. The majority of patients infected with *Wuchereria bancrofti* have:
    a. lymphedema.
    b. hydrocele.
    c. no clinical manifestations.
    d. acute adenolymphangitis (ADL).
    e. elephantiasis.
5. Which drug should not be given to patients infected with *O. volvulus* or to those infected with high-grade *Loa loa* microfilaremia?
    a. Diethylcarbamazine (DEC)
    b. Albendazole
    c. Doxycycline
    d. Azithromycin
    e. Amoxicillin

Answers

Tuberculosis

1. **b. *Mycobacterium avium-intracellulare.*** *M. bovis, M. africanum,* and *M. microti* are members of the *M. tuberculosis* complex (MTBC) and can cause TB disease. BCG is derived from *M. bovis* and can cause TB in certain individuals. Of the mycobacteria listed, *M. avium-intracellulare* is one of the many nontuberculous mycobacteria.

Table 60.1

Induration ≥ 5 mm	Induration ≥ 10 mm	Induration > 15 mm
• HIV-positive persons • Silicosis • Fibrotic changes on chest radiograph consistent with old TB • Patients with organ transplants • Immunosuppressed therapy (receiving the equivalent of >15 mg/day prednisone for ≥1 month, or tumor necrosis factor α [TNF-α] antagonists) • Recent contacts of person with TB disease • Children <5 years of age	• Immigrants from high-prevalence countries (TB incidence >20/100,000) • Residents and employees[a] of high-risk congregate settings: prisons and jails, nursing homes and other health care facilities, residential facilities for AIDS patients, and homeless shelters • Mycobacteriology laboratory personnel • Persons with clinical conditions that make them high-risk: silicosis diabetes mellitus, chronic renal failure, some hematologic disorders (e.g., leukemias and lymphomas), other specific malignancies (e.g., carcinoma of the head or neck and lung), weight loss of >10% of ideal body weight, gastrectomy, jejunoileal bypass • Injection drug users	• Persons with no risk factors for TB

HIV, Human immunodeficiency virus; *TB*, tuberculosis.

[a] For persons who are otherwise at low risk and are tested at entry into employment, a reaction of >15 mm induration is considered positive.

From American Thoracic Society and Centers for Disease Control and Prevention: Diagnostic standards and classification of tuberculosis in adults and children. *Am J Respir Crit Care Med* 161(4 Pt. 1):1376–1395, 2000.

2. **c. Prevalence of multidrug-resistant (MDR) tuberculosis cases is now approaching 12% in the United States.** Although MDR TB is concerning because of the difficulty of treatment, in 2012, the proportion of TB cases caused by MDR TB was only 1.2% in the United States.

3. **a. Hematogenous seeding.** Each of the answers is a known route of infection for the development of GU TB. However, hematogenous seeding is by far the most common one.
4. **e. Papulonecrotic tuberculid.** Papulonecrotic tuberculid is the only manifestation listed that can present early in the course of TB disease. The tuberculids are hypersensitivity reactions to MTBC antigens that were disseminated to the skin from other infectious foci, and as such, they are culture negative and typically PCR negative.
5. **e. A healthy U.S.-born teacher with a TST of 11 mm.** Refer to Table 60.1 for the Centers for Disease Control and Prevention guidelines on TST interpretation. Patients (a), (b), and (c) are likely TB infected. A BCG-vaccinated person is likely from a country with high enough incidence of TB to warrant vaccination; hence a cutoff of 10 mm is likely to apply for this person. Patient (e) has no clear risk factors for TB; hence a cutoff of 15 mm would apply for this person. However, TST cutoffs may be set differently in local public health jurisdictions according to local epidemiology.
6. d. A positive urine acid-fast bacilli (AFB) culture.
7. **d. Ethambutol.** Ethambutol is rarely hepatotoxic. Its main toxicity is ocular, such as decreased visual acuity or red-green color blindness. Streptomycin is not considered hepatotoxic either, but it is also not considered a first-line agent in the United States.
8. **c. Pyrazinamide.** By definition, MDR TB is resistant to INH, rifampin, any quinolone, and at least an additional injectable aminoglycoside. Hence, of the choices, pyrazinamide is the most likely to have efficacy against XDR TB.
9. **c. Percutaneous nephrostomy of obstructive hydronephrosis in acute renal failure.** All of the choices are appropriate indications for urological intervention. However, only (c) is emergently indicated. For the other interventions, waiting at least 4 to 6 weeks after initiation of medical therapy is preferred.
10. **a. Magnetic resonance imaging (MRI) is often used to help diagnose patients with GU TB.** Although MRI has potential uses in the diagnosis of GU TB, it is not sufficiently superior to CT or IVU to warrant its frequent use.

Schistosomiasis

1. **b. Praziquantel.** Although all of the drugs listed are antiparasitic agents, only praziquantel is used to treat schistosomiasis. In fact, praziquantel is the only drug approved for schistosomiasis by the World Health Organization (WHO).
2. **c. The cercariae.** The worm and egg stages (Fig. 60.1) are found in chronically infected humans but are intravascular or deposited in tissues such as the bladder, respectively. Cercariae infect humans by burrowing through the skin, whereupon they transform into schistosomules.
3. **d. 112 million.** Although an estimated 1 billion people are at risk of contracting schistosomiasis because they live in endemic areas, only 112 million are actively infected with *S. haematobium.*
4. **d. The egg.** The majority of human tissue pathology caused by urogenital schistosomiasis is induced by the host immune response against *S. haematobium* eggs (see Fig. 60.1). In comparison to eggs, worms, schistosomules, and cercariae are much less immunogenic and are thought to correspondingly cause much less chronic tissue pathology.
5. **a. Katayama fever.** The syndrome associated with acute schistosomiasis is named after the Katayama valley in Japan, a formerly endemic region for *Schistosoma japonicum.*
6. **e. Urine egg counts.** Although PCR and serology are highly sensitive for detecting infection, they are not considered first-line diagnostic modalities. Cystourethroscopy (Fig. 60.2) with bladder biopsy and rectal biopsy are highly invasive and reserved for difficult-to-diagnose cases or suspected cancer. Microscopic enumeration of *S. haematobium* eggs shed in urine are the diagnostic, first-line gold standard (albeit slow and impractical in many field settings).
7. **c. 3 to 5 years.** Although there have been reports that some schistosome worms can live for several decades, on average they are believed to only live for 3 to 5 years.

8. **a. Renal autotransplantation.** Renal autotransplantation is reserved for reconstruction of the urinary tract in the setting of multiple and/or large renal tumors. There are much less morbid surgical options for reconstruction of schistosomiasis-associated ureteral lesions.
9. **c. *Bulinus.*** *Biomphalaria* and *Oncomelania* are host snails for *Schistosoma mansoni* and *S. japonicum,* respectively, but not *S. haematobium. Helix* and *Achatina* snails are terrestrial and not considered hosts for human-specific schistosomes.

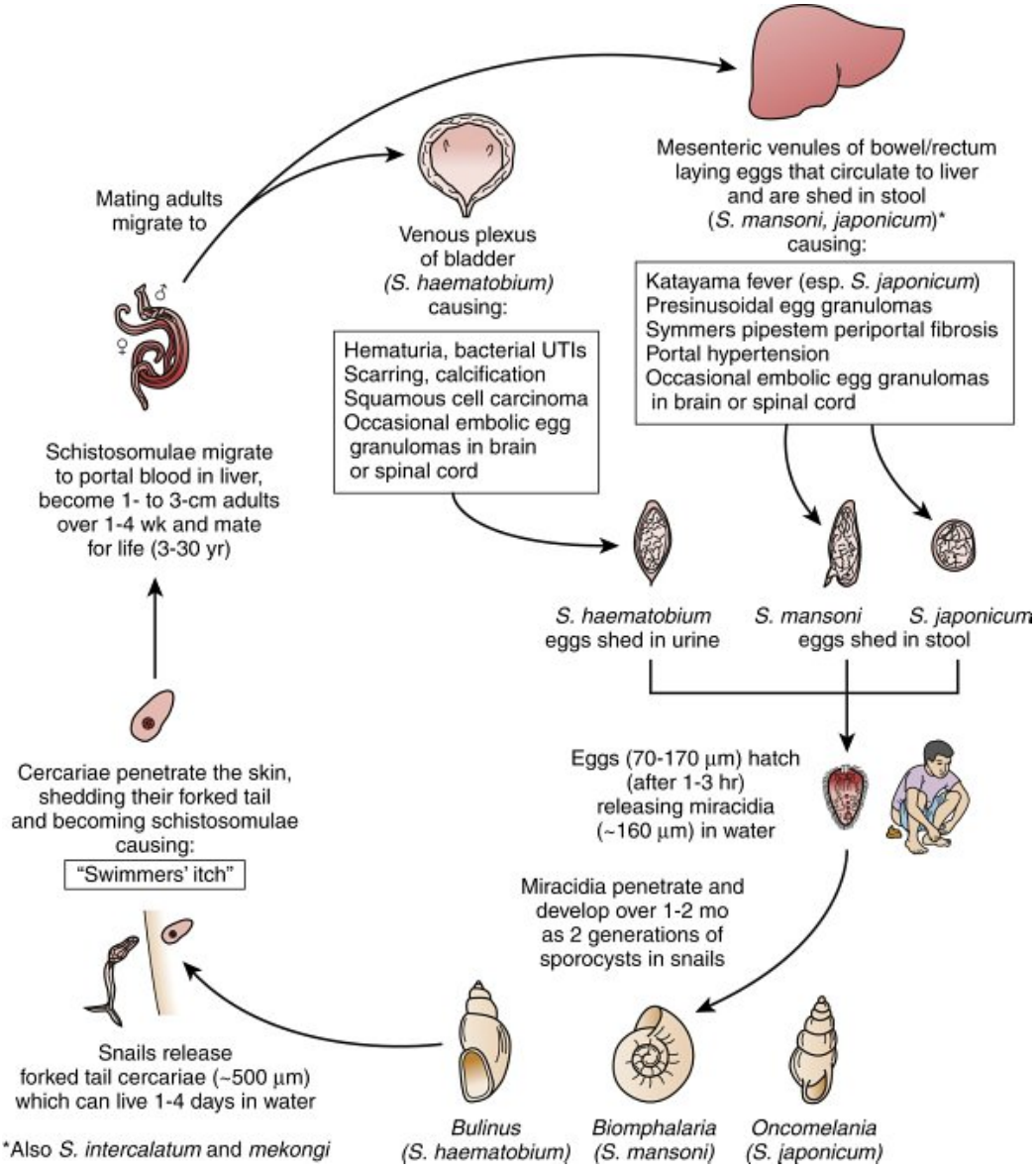


FIG. 60.1 Life cycle of a schistosome. *UTIs*, Urinary tract infections. From King CH. Schistosomiasis. In: Guerrant RL, Walker DH, Weller PF, editors. *Tropical infectious diseases, principles, pathogens, and practice*. 2nd ed. Philadelphia: Elsevier Churchill Livingstone; 2006.

Other Parasitic Infections

1. **c. *Wolbachia* spp.** *Wolbachia* endosymbionts infect *W. bancrofti*, *Brugia* spp., and *O. volvulus.* They appear to be involved in embryogenesis and, when killed with antimicrobial therapy (e.g.,

doxycycline), result in decreased microfilaria release and suppressed larval molting.

2. **c. Long-term (current) residents of endemic areas.** Because transmission is inefficient, long-term exposure to multiple infective bites appears to be necessary for transmission of LF and the development of chronic disease due to LF. Therefore, short-term visitors to endemic areas rarely develop LF, which is mostly seen in long-term residents of endemic areas.
3. **d. Sub-Saharan Africa.** Although endemic to Latin America and the Middle East as well, 99% of persons who have onchocerciasis live in sub-Saharan Africa.
4. **c. No clinical manifestations.** Although *W. bancrofti* infection can lead to all of these clinical manifestations, most infected persons remain asymptomatic.
5. **a. Diethylcarbamazine (DEC).** DEC can cause blindness in patients with onchocerciasis (due to the inflammatory response to parasites in the anterior chamber of the eye) and encephalopathy in patients with high-grade *L. loa* microfilaremia.

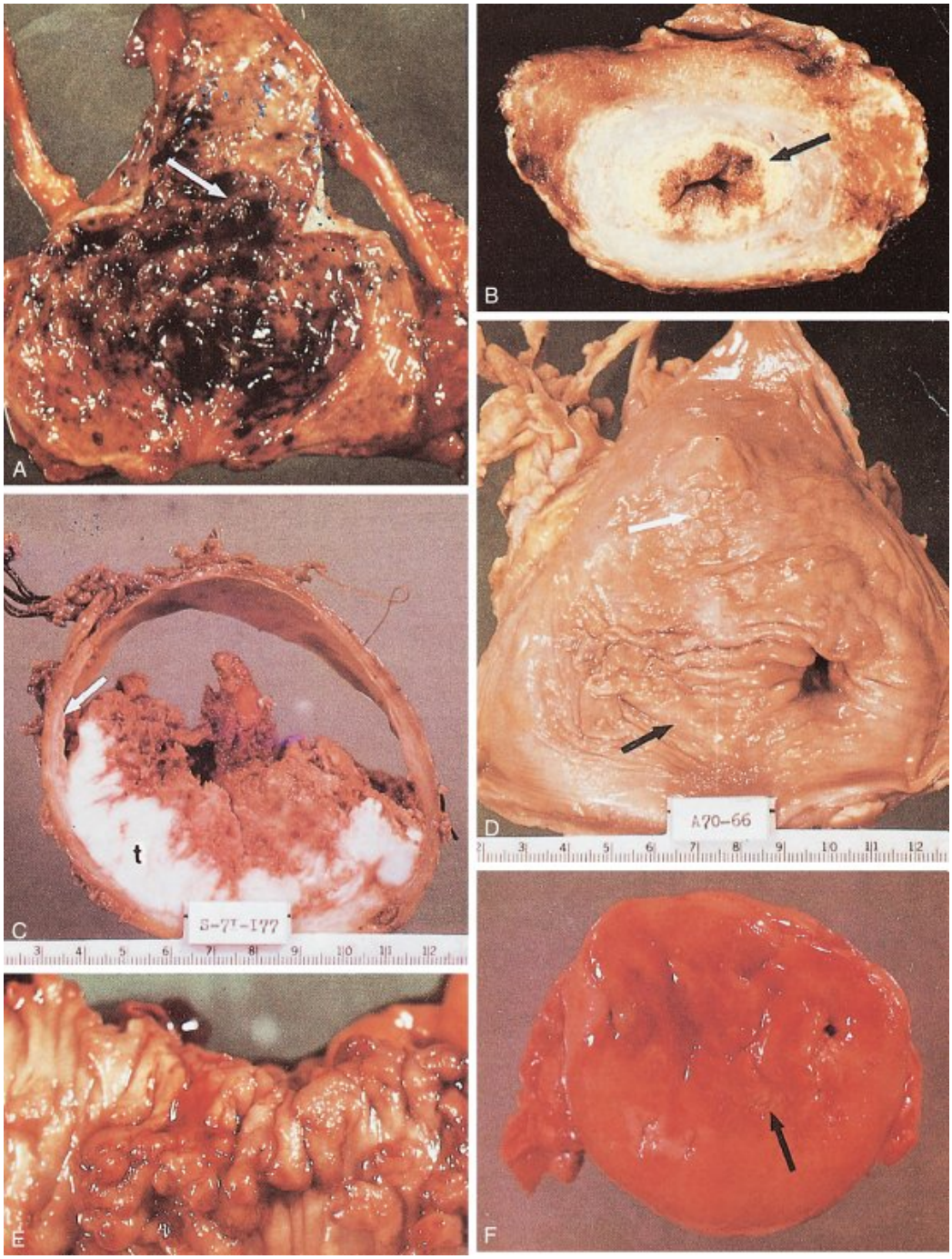


FIG. 60.2 Macroscopic appearance of human urinary schistosomiasis. (A) Urinary bladder opened with an anterior Y incision. The posterior and apical walls have many erythematous, granular, sessile, and pedunculated polyps *(arrow)*, characteristic of the early active stage of urinary schistosomiasis. (B) Coronal section through the apex of a formalin-fixed urinary bladder. The lamina propria has been expanded and is replaced by a yellow-tan, finely granular, sandy patch *(arrow)*, which is characteristic of chronic inactive foci. Small sandy patches are sprinkled through the fibrotic,

atrophic detrusor muscle, even in perivesical fat. The more superficial erythematous portion of the lamina propria contains some viable eggs with granulomatous response (chronic active stage of urinary schistosomiasis). (C) Coronal section through the middle of a urinary bladder after formalin inflation and fixation. The lamina propria *(arrow)* has been replaced by a concentric sandy patch, most prominent at the margin of the exophytic, moderately differentiated squamous cell carcinoma. The bladder wall is attenuated except for the tumor *(t)*. No evidence of recent oviposition was found in the lower urinary tract (chronic inactive stage of urinary schistosomiasis, usually found with the bilharzial bladder cancer syndrome). (D) Urinary bladder opened with anterior Y incision shows several features of severe chronic inactive urinary schistosomiasis. The entire lamina propria has been replaced by a sandy patch. Foci of epidermization are seen at or near the white arrow. The left ureteral orifice *(right)* is markedly dilated (the so-called golf-hole ureter of schistosomal uropathy). The right ureteral orifice *(point of black arrow)* is markedly stenotic. (E) Rectosigmoid colon with polyposis. Numerous sessile and pedunculated polyps are seen. Many are erythematous, indicative of active oviposition with granuloma formation. Some have necrotic hemorrhagic tips. (F) Mucosal surface of partial cystectomy specimen (4- to 5-cm ellipse) from a patient with the chronic inactive stage of the disease. There is a stellate chronic schistosomal ulcer. Despite the inactivity of the disease, these ulcers may bleed profusely. Pale mucoid flecks at the margin of the ulcer *(arrow)* are areas of adenoid (goblet cell) metaplasia.

Chapter Review

1. Untreated genitourinary TB can lead to irreparable tissue damage and extensive fibrosis, with serious consequences such as renal failure and infertility.
2. The principal route by which genitourinary TB develops is via hematogenous spread, localized to only the genito-urinary tract or disseminated to multiple organ systems.
3. Genitourinary TB is associated with nonspecific symptoms and can be indolent, with fewer than 20% presenting with constitutional symptoms.
4. Eighty percent of genitourinary TB occurs in the kidney, and typical laboratory findings include sterile pyuria or hematuria.

5. Ureteral TB can lead to strictures throughout the ureter and impart a pipestem or beaded corkscrew appearance on imaging; bladder TB can lead to severe contracture and is a late complication more common in developing countries.
6. Another common presentation of genitourinary TB is chronic prostatitis that persists despite multiple course of antibiotics.
7. The gold standard for diagnosis of genitourinary TB is a positive AFB culture from urine or tissue biopsy. First void urine is the best sample for culture.
8. Genitourinary TB can also be made with alternate methods of diagnosis such as nucleic acid amplification of *M. tuberculosis* or the presence of granulomas on histopathology examination of tissue.
9. Genitourinary TB generates a wide spectrum of imaging findings. The test of choice depends on disease location. All patients should also have a chest x-ray to exclude concomitant infectious pulmonary TB.
10. Combination therapy with first-line antituberculous drugs achieves the best cure rates in the shortest time frame and should start with isoniazid, rifampin, pyrazinamide, and ethambutol.
11. Surgical interventions are performed to relieve urinary obstruction, to drain infected material, to remove nonworking infected kidneys in cases resisting cure, to improve medically resistant hypertension secondary to a functionally excluded kidney, or to reconstruct the urinary tract.
12. The optimal timing of surgery is 4 to 6 weeks after the initiation of therapy. This delay allows active inflammation to subside, the bacillary load to decrease, and lesions to stabilize.
13. Monitoring for relapse is important after treatment of genitourinary TB because it can occur in up to 22% of cases.
14. Rifampin resistance serves as a surrogate marker for MDR TB.
15. *S. haematobium* has a terminal spine and dwells principally in the perivesical venous plexuses.
16. Schistosomiasis may cause inflammatory polyps of the bladder, sandy spots in the bladder (which represent submucosal egg deposition), calcification of the entire outline of the bladder, and

strictures of the ureter (usually in the distal portion) with hydronephrosis. It may be associated with bladder cancer.

17. Squamous cell carcinoma of the bladder is the most common histologic variant occurring as a result of schistosomiasis. These cancers are usually well differentiated or verrucous and therefore carry an overall good prognosis.
18. *W. bancrofti* accounts for 90% of human LF.
19. Obstructive lymphatic disease typically occurs in people who have multiple reinfections following the initial infection with filaria.
20. *W. bancrofti* results in chyluria and filarial hydrocele and occasional extensive scrotal and penile lymphedema.
21. Echinococcosis may result in cysts in the kidney; cyst rupture or spillage during surgical removal can cause anaphylaxis.

PART V
Molecular and Cellular Biology

OUTLINE

61. Basic Principles of Immunology and Immunotherapy in Urologic Oncology
62. Molecular Genetics and Cancer Biology

61

Basic Principles of Immunology and Immunotherapy in Urologic Oncology

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Keywords

BCG; CD4; CD8; CTLA-4; immunotherapy; lymphocyte; neutrophil; NK cell; PD-1; PD-L1; T cell; Treg

Questions

1. The immune system has two basic arms, designated “innate” and adaptive. Regarding these divisions, which of the following is true?
    a. The adaptive immune system is the more primitive of the two.
    b. The innate immune system includes long-lived “memory” cells that make secondary responses to pathogens more rapid and stronger.
    c. The adaptive immune system is the only part of the immune system involved in the immune response to cancer.
    d. The innate immune response is driven by recognition of repeated patterns in pathogens called pathogen associated molecular patterns (PAMPs).
2. Immunotherapy for non–muscle-invasive bladder cancer is based on intravesical administration of bacillus Calmette-Guérin (BCG).

Initial contact between BCG and the immune system involves which cell type?

a. Neutrophils, also known as polymorphonuclear cells (PMNs)
b. CD4+ (helper) T cells
c. Macrophages (“Big eaters”)
d. CD8+ (killer) T cells
e. CD8+ regulatory T cells (Tregs)

3. Macrophages and neutrophils have the following feature(s) in common:
    a. they can both ingest pathogens.
    b. they can both present peptide antigens from pathogens on major histocompatibility complex (MHC) class II molecules.
    c. they can both kill pathogens via oxidative processes.
    d. they can both migrate along chemokine gradients to sites of infection.
    e. c and d.
4. Cytokines are chemical messengers that direct immune cell migration and differentiation. Which of the following is true regarding cytokines?
    a. Interleukin 8 (IL-8) is a Th1 cytokine that helps naive CD4+ T cells differentiate into cells that secrete IL-4.
    b. Interferon α is an adaptive cytokine that has no role in the response to BCG.
    c. Interferon γ is an innate cytokine mostly secreted by luminal epithelial cells in the bladder.
    d. Th1 cytokines include IL-4, IL-5, and IL-10.
    e. The primary Th1 cytokine is IL-12.
5. The most important adaptive immune cells are B and T cells. Which of the following is true?
    a. B cells are called B cells because they come from the bone marrow.
    b. CD4 (helper) T cells are mostly of a single phenotype.
    c. CD8 T cells are specialized for cytotoxicity; they evolved to destroy virally infected cells.
    d. Regulatory T cells (Tregs) are a subset of CD8 T cells.

e. In non–muscle-invasive bladder cancer, a Th2 response involving IL-4, IL-5, and IL-10 is associated with a better outcome.

6. The major function of B cells is to:
    a. kill bacterially infected cells.
    b. generate antibodies which help macrophages to phagocytose and destroy pathogens.
    c. differentiate into memory cells that provide a secondary immune response about equal in magnitude to a primary response.
    d. ingest pathogens.
    e. augment the function of BCG by generating anti-BCG antibodies that lead to BCG persistence in the bladder.
7. CD4 T cell subsets include:
    a. Tregs that augment an antitumor immune response.
    b. Th1 T cells that secrete IL-1.
    c. Th2 T cells that secrete IL-4, IL-5, IL-10, and IL-13.
    d. Th9 T cells that are involved in CD8 T-cell killing.
    e. Th8 T cells that have a cytotoxic function.
8. The immune editing hypothesis explains how tumors evolve in an immunologically competent host. The three E's in the immune editing hypothesis (in order) include:
    a. elimination, equilibrium, and escape.
    b. escape, equilibrium, and elimination.
    c. elucidation, escape, and election.
    d. evolution, equilibrium, and elimination.
9. There is a clear epidemiologic link between chronic inflammation and certain genitourinary (GU) cancers. Which of the following best describes this relationship?
    a. Correlative data are strongest in kidney cancer, where chronic inflammation leads to clear cell cancer or renal cell carcinoma.
    b. Data for this link are absent in prostate cancer.
    c. In bladder cancer, chronic infection with *Schistosoma haematobium* (schistosomiasis) rarely leads to cancer with a squamous cell phenotype.

d. In prostate cancer, inflammation has been associated with the consumption of charred meat and its associated toxins.

10. The presence of CD8 T cells is associated with:
    a. a better prognosis in all GU tumor types, since CD8 T cells are cancer-fighting cytotoxic cells.
    b. a better prognosis in bladder cancer and prostate cancer.
    c. A worse prognosis in kidney cancer.
    d. A better prognosis in prostate cancer. This is well documented, since prostate tumors are frequently highly infiltrated with CD8 T cells.
11. Immune checkpoint molecules are:
    a. a series of intracellular proteins that regulate progression of T cells through the cell cycle.
    b. a series of extracellular proteins that promote T-cell activation and effector function.
    c. a series of proteins on innate immune cells that inhibit their function.
    d. a series of cell-surface proteins that attenuate CD8 T-cell effector function when bound to their specific cognate ligands.
12. In non–muscle-invasive bladder cancer (NMIBC), the immunotherapeutic effects of BGC are mediated by all of the following EXCEPT which?
    a. The innate immune system, mostly through the activation of neutrophils and macrophages
    b. The release of chemokines from luminal epithelial cells in the bladder
    c. CD4 T cells of the Th2 phenotype
    d. CD8 T cells of the Tc1 phenotype that participate in the direct killing of tumor cells
13. Cancer vaccines are intended to raise an adaptive immune response to a target antigen expressed preferentially on tumor cells. Which of the following is a US Food and Drug Administration (FDA)–approved vaccine for a GU cancer type?
    a. ProstVac VF, which is used in prostate cancer
    b. IMA901, a multi-peptide vaccine for kidney cancer
    c. Gardasil, a vaccine against human papillomavirus

d. Sipuleucel-T, a prostatic acid phosphatase (PAP)–directed vaccine for prostate cancer

14. The interaction between the immune checkpoint molecule PD-1 (on CD8 T cells) and its ligand PD-L1 (on tumor cells) does which of the following?
    a. Helps CD8 T cells to activate dendritic cells
    b. Drives CD8 T cells to secrete interferon γ, promoting antitumor responses
    c. Mediates an "exhausted" nonfunctional phenotype in the CD8 T cells that express PD-1
    d. Signals through PD-L1 back into the tumor cell to promote tumor cell destruction
15. In bladder cancer, blockage of the interaction between PD-1 and PD-L1 using monoclonal antibodies:
    a. provides a survival benefit but only when used with chemotherapy.
    b. improves both progression-free and overall survival as compared with second-line chemotherapy.
    c. improves survival but leads to tumor response (shrinkage) in less than 10% of treated patients.
    d. signals through PD-L1 back into the tumor cell to promote tumor cell destruction.
    e. frequently leads to complete responses.
16. PD-1 is not the only immune checkpoint molecule expressed on T cells in tumors; other checkpoint molecules include CTLA-4, LAG-4, and others. Simultaneously blocking *two* immune checkpoints (PD-1 and CTLA-4) in first-line kidney cancer:
    a. is less toxic than blocking a single immune checkpoint molecule.
    b. is more effective than first-line therapy with a vascular endothelial growth factor (VEGF) TKI (a tyrosine kinase inhibitor).
    c. never leads to complete responses.
    d. most likely functions by blocking both molecules on the same cell type.
    e. rarely leads to autoimmunity in endocrine organs.

Answers

1. **d. The innate immune response is driven by recognition of repeated patterns in pathogens called pathogen associated molecular patterns (PAMPs). The innate system is more primitive and does not contain cell types that can give rise to memory cells.** In most immune responses, the innate and adaptive systems work together.
2. **c. Macrophages ("Big eaters"). Although BCG does bind directly to bladder epithelium, the first immune cell involved in BCG responses is the macrophage ("big eater").**
3. **e. c and d. Macrophages and PMNs are both innate immune cells involved in the early phases of an immune response.** Although both have potent cytotoxic activity, only macrophages can ingest pathogens and present antigens in the context of MHC molecules.
4. **e. The primary Th1 cytokine is IL-12. Th1 responses are characterized by IL-12, interferon γ (not α) and tumor necrosis factor (TNF) α.** Th1 responses are involved in viral clearance and in antitumor immune responses. Th2 responses (IL4, 5, 10, and 13) are associated with chronic inflammation and infection and generally prevent an antitumor immune response. In bladder cancer, Th2 cytokines in the urine after BCG treatment portend a generally worse outcome.
5. **c. CD8 T cells are specialized for cytotoxicity; they evolved to destroy virally infected cells. CD4 T cells are complex and include a number of phenotypes (Th1, Th2, Tregs, and others).** CD8 T cells are cytotoxic and are capable of killing nearly any target cell (including cells resistant to chemotherapy). Although B cells DO come from the bone marrow, that is not how they got their name. They were discovered in chickens in an organ called the bursa of Fabricius. The B comes from the word *bursa.*
6. **b. Generate antibodies which help macrophages to phagocytose and destroy pathogens. B cells secrete antibodies, which help to clear pathogens.** They also differentiate into memory B cells,

which drive a stronger and faster secondary response on reexposure to a pathogen. The generation of antibodies to BCG is probably a negative event; it would be expected to lead to a faster clearance and a less robust antitumor effect.

7. **c. Th2 T cells that secrete IL-4, IL-5, IL-10, and IL-13. As earlier, Th2 cells secrete IL4, 5, 10, and 13, all of which generally inhibit a successful antitumor response.**
8. **a. Elimination, equilibrium and escape. See text for explanation.**
9. **d. In prostate cancer, inflammation has been associated with the consumption of charred meat and its associated toxins. In GU cancers, the strongest linkage between chronic inflammation and tumorigenesis is in bladder cancer, where chronic infection with schistosomiasis leads to squamous cell cancers.** There are also strong data in prostate cancer, although the precise etiology of prostatic inflammation has yet to be fully defined.
10. **c. A worse prognosis in kidney cancer. Although cytotoxic CD8 T cells are generally associated with a better prognosis, in kidney cancer the opposite relationship has been described.** This is a puzzling finding, which is not well understood. In bladder cancer, a robust CD8 infiltrate is generally associated with an improved outcome.
11. **d. A series of cell-surface proteins that attenuate CD8 T-cell effector function when bound to their specific cognate ligands. Immune checkpoint molecules are cell-surface proteins that attenuate CD8 T cell function.**
12. **c. CD4 T cells of the Th2 phenotype. As earlier, Th2 CD4 T cells are associated with a worse outcome in bladder cancer.**
13. **d. Sipuleucel-T, a prostatic acid phosphatase (PAP)–directed vaccine for prostate cancer. Although many cancer vaccines have been tested in GU cancers, only Sipuleucel-T is approved.** This vaccine targets the antigen PAP in men with asymptomatic or minimally symptomatic metastatic castration-resistant prostate cancer.
14. **c. Mediates an "exhausted" nonfunctional phenotype in the CD8 T cells that express PD-1. Expression of PD-1 on tumor-infiltrating CD8 T cells is associated with a nonfunctional,**

exhausted phenotype. Blockage of the PD-1/PD-L1 interaction with monoclonal antibodies restores CD8 T-cell function, leading to tumor destruction (and shrinkage) in a minority of treated patients.

15. **b. Improves both progression-free and overall survival as compared with second-line chemotherapy. Blockage of either PD-1 or PD-L1 is effective in bladder cancer in the second-line setting (as compared with chemotherapy).** The overall response rate to these agents is in the range of 20% to 25%.
16. **b. Is more effective than first-line therapy with a vascular endothelial growth factor (VEGF) TKI (a tyrosine kinase inhibitor). Combined immune checkpoint blockade with anti-PD-1 (nivolumab) and anti-CTLA-4 (ipilimumab) is an approved first-line therapy for clear-cell RCC.** It is more effective than treatment with the TKI sunitinib, as evidence by improved overall survival and progression-free survival. However, it is also considerably more toxic, with endocrine-related adverse events being reasonably common.

Chapter Review

1. An immune response begins with an innate response, which is rapid but relatively nonspecific. It then progresses to include the adaptive immune system, which is characterized by both specificity and memory.
2. For an antitumor immune response, a Th1 response dominated by interferon γ, IL-2, and TNF α is desired.
3. CD4+ regulatory T cells (Tregs) inhibit an adaptive immune response.
4. The immune editing hypothesis explains how early tumors can be recognized and eliminated by the immune system, whereas clinically evident tumors must escape immune recognition to evolve.
5. Bladder cancer is promoted by chronic inflammation initiated by infection or other stimuli.

6. In kidney cancer, there is good evidence for an ongoing adaptive, CD8 T-cell–mediated response, but paradoxically, an increased CD8 density is associated with a worse prognosis.
7. Prostate cancer may also be driven by chronic inflammation.
8. Expression of immune checkpoint molecules on tumor-infiltrating lymphocytes (TILs) attenuates the adaptive immune response to GU tumors.
9. BCG therapy for bladder cancer stimulates both the innate and adaptive immune systems and is a prototype for successful cancer immunotherapy.
10. Cancer vaccines have been evaluated in both kidney and prostate cancer. A single vaccine (Sipuleucel-T) is FDA-approved for metastatic prostate cancer.
11. Several other vaccine approaches in phase III trials in kidney and prostate cancer have failed.
12. BCG therapy for bladder cancer stimulates both the innate and adaptive immune systems and is a prototype for successful cancer immunotherapy.
13. Cancer vaccines have been evaluated in both kidney and prostate cancer.
14. A single vaccine (Sipuleucel-T) is FDA-approved for metastatic prostate cancer.
15. Cancer vaccines are generally well tolerated but result in few objective clinical responses.
16. Blocking the PD-1/PD-L1 interaction is effective in second-line bladder cancer and has activity in first-line patients who are cisplatin-ineligible.
17. PD-L1 expression may distinguish bladder cancer patients more likely to respond to anti-PD-1/anti-PD-L1 therapy.
18. Anti-PD-1 is effective in second-line kidney cancer.
19. Anti-PD-1/PD-L1 has not shown robust monotherapy activity in metastatic prostate cancer.

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Molecular Genetics and Cancer Biology

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Keywords

cancer; molecular genetics; tumor suppressor genes

Questions

1. Deoxyribonucleic acid (DNA) is composed of all of the following elements EXCEPT:
    a. a base, either a purine or a pyrimidine.
    b. a sugar, called ribose.
    c. a phosphate.
    d. two complementary strands.
    e. hydrogen bonds.
2. The physical chemistry of DNA bases requires:
    a. uracil to form hydrogen bonds with guanine.
    b. purine to form hydrogen bonds with another purine.
    c. pyrimidine to form hydrogen bonds with a purine.
    d. adenine to form hydrogen bonds with cytosine.
    e. thymine to form hydrogen bonds with guanine.
3. Which of the following does DNA gene expression require?
    a. Linear DNA to be converted into linear ribonucleic acid (RNA), a process called translation

b. Conversion of linear RNA into a linear set of amino acids, a process called transcription
c. Protein synthesis exclusively within the nucleus
d. Mitosis
e. A mechanism to bridge the gap between the genetic code and protein synthesis

4. Which of the following statements about transcriptional regulation is NOT true?
    a. Two general components involved in transcriptional regulation are specific sequences in the RNA and proteins that interact with those sequences.
    b. In addition to the genetic information carried within the nucleotide sequence of DNA, it provides specific docking sites for proteins that enhance the activity of the transcriptional machinery.
    c. Specific sequences within the promoter or enhancer region of a gene are called response elements.
    d. DNA sequences, often referred to as consensus sequences, are found in many genes and respond in a coordinated manner to a specific signal.
    e. It is an important mechanism to ensure coordinated gene expression.
5. What is alternative splicing?
    a. A form of protein modification occurring after a mature polypeptide is produced
    b. A modification to DNA during meiosis
    c. A process of including or excluding certain exons in an mRNA transcript
    d. A process in which novel RNA sequences are randomly inserted into a transcript
    e. A method of RNA degradation
6. Which of the following statements is TRUE regarding the nuclear matrix?
    a. It has the same protein composition in every tissue type.
    b. It is the site of mRNA transcription.

c. It has the same protein composition whether a cell is proliferating or undergoing differentiation.
d. It provides a mechanism to trace the cell type of origin for a cancer, because the nuclear matrix is identical within tissue types.
e. It is a form of DNA.

7. What happens in the process of translation?
   a. The RNA message of four parts (the nucleotides A, U, C, and G) is converted into 20 amino acids by using a functional group of three adjacent nucleotides called a codon.
   b. Each amino acid is encoded by only one codon.
   c. Shifts in the reading frame are of no consequence in the production of the polypeptide chain because of the fidelity of template DNA.
   d. Single-base substitutions always encode for the identical amino acid, known as a polymorphism.
   e. Transfer of genetic information from the DNA to the RNA occurs.
8. Which of the following statements about ubiquitization is NOT true?
   a. Ubiquitization is an important regulatory mechanism of a cell, used in the efficient disposal of proteins.
   b. A small protein called ubiquitin is linked to a protein, tagging it for destruction.
   c. The proteosome is the site of protein-ubiquitin complex degradation.
   d. The proteosome has a cylindrical shape.
   e. The targeted inhibition of proteosome function appears to enhance cancer progression.
9. Which of the following statements about oncogenes is TRUE?
   a. They are mutated forms of abnormal genes, known as proto-oncogenes.
   b. They can be produced by an inactivating mutation of a proto-oncogene, resulting in the silencing of the gene.
   c. They can be produced by gene amplification, resulting in many copies of the gene, or by chromosomal rearrangement.

d. They are always due to retroviruses, capable of inducing malignant transformation of normal cells.
e. They are endogenous cancer-fighting genes.

10. Which of the following statements about *hyper*methylation is TRUE?
    a. It is a direct change to the DNA sequence, similar to a mutation that alters the normal base-pairing.
    b. It occurs exclusively on cytosine nucleotides in the dinucleotide sequence CG.
    c. Somatic methylation of CpG dinucleotides in the regulatory regions of genes is very often associated with increased transcriptional activity, leading to increased expression of that gene.
    d. In cancer, hypermethylation is associated with enhanced activity of oncogenes, and demethylation may be an effective strategy for the treatment of cancer.
    e. It marks cells for ubiquitinization.
11. Von Hippel-Lindau disease predisposes patients to:
    a. epididymal carcinoma
    b. clear cell renal carcinoma
    c. papillary renal cell carcinoma
    d. adrenocortical carcinoma
    e. all of the above
12. Hereditary prostate cancer genes that have been proven to cause prostate carcinoma include:
    a. ELAC2 (elaC ribonuclease Z 2)
    b. MSR1 (macrophage scavenger receptor 1)
    c. RNASEL (ribonuclease L)
    d. none of the above
    e. all of the above
13. Which of the following hereditary tumor syndromes are associated with genitourinary tumors?
    a. Von-Hippel-Lindau syndrome
    b. Birt-Hogg-Dubé syndrome
    c. Beckwith-Wiedemann syndrome
    d. None of the above

e. All of the above

14. Which of the following cancer-associated chromosomal abnormalities would be most likely to be associated with inactivation of a tumor suppressor gene?
    a. Inversion
    b. Tetraploidy
    c. Amplification
    d. Deletion
    e. Double Minutes
15. The two main points of control in the cell cycle are:
    a. S and G_0
    b. S and the G_2M boundary
    c. M and the G_1S boundary
    d. G_1S and G_2M boundaries
    e. G_2 and M
16. The TP53 tumor suppressor gene plays a critical role in which of the following processes?
    a. Apoptosis
    b. Angiogenesis
    c. DNA replication
    d. Signal transduction
    e. All of the above
17. INK4 family members inhibit the activity of:
    a. cyclin D/cdk4.
    b. cyclin E/cdk2.
    c. cyclin A/cdk2.
    d. cyclin B/cdc2.
    e. all of the above.
18. Cyclin-cdk complexes primarily function at the G_1S boundary by:
    a. dephosphorylation of RB1.
    b. phosphorylation of MDM2.
    c. dephosphorylation of E2F.
    d. phosphorylation of E2F.
    e. phosphorylation of RB1.

19. The regulatory proteins at the G_2M checkpoint primarily respond to:
    a. hypoxia.
    b. nutrient poor environment.
    c. DNA damage.
    d. cytokines.
    e. all of the above.
20. Nucleotide excision repair (NER) primarily protects the cell from DNA damage caused by:
    a. reactive oxygen species.
    b. DNA polymerase errors.
    c. double-stranded breaks.
    d. ultraviolet light.
    e. all of the above.
21. Which of the following repair pathways is responsible for repairing double-strand DNA breaks?
    a. MSH2/MSH6
    b. homologous recombination (HR)
    c. NER
    d. mismatch repair (MMR)
    e. none of the above
22. Which of the following genes has been linked to double-strand break repair?
    a. TP53
    b. VHL (von Hippel-Lindau tumor suppressor)
    c. BRCA1 (breast cancer type 1 susceptibility protein)
    d. RB1 (retinoblastoma 1)
    e. PTEN (phosphatase and tensin homolog)
23. Procaspases are activated by which of the following?
    a. Phosphorylation
    b. Ubiquitination
    c. Dimerization
    d. Mitochondrial import
    e. Proteolytic cleavage

24. Ligand-dependent apoptosis is an attractive therapeutic target because activation is:
    a. independent of TP53.
    b. dependent on TP53.
    c. independent of caspases.
    d. dependent on caspases.
    e. dependent on RB1.
25. The TP53-induced apoptosis is mediated through:
    a. APAF-1/caspase 9.
    b. CD95 receptor.
    c. TRAIL.
    d. Bcl-2.
    e. CDKN1A ($p21^{cip}$).
26. How do proapoptotic bcl-2 family members function? By:
    a. digesting the mitochondria.
    b. increasing the cellular membrane permeability.
    c. increasing the mitochondrial membrane permeability.
    d. directly activating executioner caspases.
    e. increasing nuclear membrane permeability.
27. The enzyme telomerase immortalizes cells by:
    a. protecting the cells from DNA damage.
    b. stabilizing TP53.
    c. allowing the cell to grow in a nutrient poor environment.
    d. inhibiting apoptosis.
    e. maintaining chromosomal length.
28. Telomere loss can lead to all of the following EXCEPT:
    a. irreversible cell cycle exit termed senescence.
    b. apoptosis.
    c. DNA hypomethylation.
    d. chromosomal instability.
    e. increased tumor initiation.
29. Which of the following chromosomal rearrangements is not typically associated with a genitourinary malignancy?
    a. Fusion of BCR to ABL via chromosome translocation
    b. Fusion of TMPRSS2 to ERG via intra-chromosomal deletion

c. Fusion of MiTF/TFE gene family members via chromosome translocation
d. Isochromosome 8q
e. Loss of chromosome 9

30. An isochromosome of 12p has been identified in which genitourinary carcinoma?
    a. Testis
    b. Prostate
    c. Renal
    d. Bladder
    e. Penile
31. Which of the following has been most consistently implicated in prostate cancer risk from GWAS studies?
    a. The 8q24 chromosomal region
    b. MSR1
    c. RNASEL
    d. TMPRSS2-ERG fusions
    e. None of the above
32. What type of genetic alteration in cancer could be identified using next-generation RNA sequencing (RNA-seq)?
    a. MicroRNAs associated with cancer
    b. Non-coding RNAs associated with cancer
    c. Gene fusions
    d. mRNA overexpression in cancer
    e. All of the above
33. An advantage of DNA sequencing-based technologies over standard clinical microbial culture for pathogen identification in body sites such as the urinary tract is:
    a. sequencing techniques can be used to define bacterial growth conditions
    b. sequencing technologies can identify the presence of organisms that cannot be grown in culture
    c. microbial culture cannot assist with the classification of bacterial species
    d. microbial culture cannot accurately identify pathogens
    e. none of the above

34. Which of the following medications has been shown to improve overall survival with fewer severe side effects among patients with clear cell renal cell carcinoma who received previous treatment?
    a. Atezolizumab
    b. Pembrolizumab
    c. Everolimus
    d. Nivolumab
    e. None of the above

Answers

1. **b. A sugar, called ribose.** Ribose is an element of RNA, not DNA. In a rudimentary form, DNA is the fusion of three different elements: a base (either a pyrimidine or a purine), a sugar (in the case of DNA called 2-deoxyribose; for RNA called ribose), and a phosphate (which links individual nucleotides together). The repeating connections between the phosphates and the sugars provide the backbone from which the information-carrying bases protrude. In its "resting" or non-replicating form, this chain of elements forms a helix of two complementary strands; this double helix is held together by the hydrogen bonds.
2. **c. A pyrimidine to form hydrogen bonds with a purine.** The double helix is held together by the formation of hydrogen bonds between the pyrimidine on one strand and the purine base on the other. Uracil is in RNA not DNA. Purines do not form bases with purines. The purine adenine (A) always forms two hydrogen bonds to the pyrimidine thymine (T), and the purine guanine (G) always forms three hydrogen bonds to the pyrimidine cytosine (C), one consequence being that A-T bonding is weaker than G-C bonding.
3. **e. A mechanism to bridge the gap between the genetic code and protein synthesis.** The physical locations of DNA and its genetic code and protein synthesis and the manifestation of that code are separate: DNA is in the nucleus, and protein synthesis is cytoplasmic. The DNA message must be converted into mRNA by a process called transcription in the nucleus. The mRNA is

transferred into the cytoplasm where the mRNA is converted into a protein by a process called translation.

4. **a. Two general components involved in transcriptional regulation are specific sequences in the RNA and proteins that interact with those sequences.** Specific sequences in the DNA called response elements bind to the nuclear proteins that control transcription. Importantly, this provides a mechanism to coordinate gene expression by providing similar docking sites for genes that need to be expressed at specific time points. Whereas mRNA transcription, stability, transport, and translation are all highly regulated, they are not controlled by the transcriptional machinery binding to specific mRNA sequences.
5. **c. A process of including or excluding certain exons in an mRNA transcript.** A specific gene may have multiple similar (not identical) forms (isoforms). This is accomplished by having specific exons included or excluded from the final mRNA transcript. This allows one DNA sequence to produce several protein products that have different functions. It is not a random process but is very tightly regulated to ensure that the correct mRNA transcript is produced in the correct cell at the correct point in time.
6. **b. It is the site of mRNA transcription.**
7. **a. The RNA message of four parts (the nucleotides A, U, C, and G) is converted into 20 amino acids by using a functional group of three adjacent nucleotides called a codon.** There is significant redundancy with several codons encoding for each amino acid. As a result, the RNA sequence between individuals can be different and yet still encode the same amino acid sequence (a polymorphism). However, some polymorphisms do result in amino acid changes. Shifts in reading frame are the most deleterious mutations because they result in a change in all the codons after the insertion/deletion and as a result a dramatic change in the amino acid sequence.
8. **e. The targeted inhibition of proteosome function appears to enhance cancer progression.** Degradation of proteins in the cell is an active, not passive, process. Ubiquitinization is the process by

which proteins are tagged for transport to the proteosome for destruction. Perturbation of this process is often found in cancer, and therefore inhibition of ubiquitinization is a promising therapy for cancer.

9. **c. They can be produced by gene amplification, resulting in many copies of the gene, or by chromosomal rearrangement.** There are at least three ways a proto-oncogene can be converted into an oncogene. First, a mutation can occur within the coding sequence, producing a permanently activated form of the gene. A second mechanism converting a proto-oncogene into an oncogene is through gene amplification. A third mechanism of oncogene formation is through chromosomal rearrangement.
10. **b. It occurs exclusively on cytosine nucleotides in the dinucleotide sequence CG.** Hypermethylation is a normal process by which DNA is modified by the addition of a methyl group to a cytosine nucleotide in a CpG DNA sequence. There is no change in DNA sequence. Methylation results in gene silencing, that is, decreased expression of the gene. As a result, hypermethylation in cancer is associated with decreased transcription and, therefore, reduces expression of tumor suppressor genes. Ubiquitinization decreases levels of proteins but does so by increasing the degradation of the protein. Whereas alterations in ubiquitinization occur in cancer, they are not directly related to methylation.
11. **b. Clear cell renal carcinoma.** VHL is a hereditary tumor syndrome that predisposes patients to clear cell renal carcinoma, retinal angiomas, pheochromocytomas, hemangiomas of the CNS, epididymal cystadenomas, and pancreatic islet cell tumors. It is not associated with epididymal, papillary renal cell, or adrenocortical carcinomas.
12. **d. None of the above.** All of the genes above have been linked to prostate cancer by both linkage studies examining families with a strong predisposition to prostate cancer and in some cases by case control studies examining polymorphisms within these genes. However, no data have conclusively demonstrated that these genes cause prostate cancer.

13. **e. All of the above.** Each of these syndromes includes increased risk of specific genitourinary malignancies among their spectra of pathologies.
14. **d. Deletion.** A common mechanism of tumor suppressor gene inactivation is through deletion of the gene or a chromosomal region containing the gene. The other abnormalities listed produce either no net loss of genetic material or lead to a gain of genetic material, a finding often associated with increased oncogene activity due to increased copy number of the oncogene.
15. d. **G_1S and G_2M boundaries.**
16. **a. Apoptosis.** Active TP53 binds to the promoter region of TP53-responsive genes and stimulates the transcription of genes responsible for cell cycle arrest, repair of DNA damage, and apoptosis. TP53 responds to DNA damage by inducing cell cycle arrest through CDKN1A (p21cip1) and then transcriptionally activating DNA repair enzymes. If the cell cannot arrest growth and/or repair the DNA, TP53 induces apoptosis. Whereas angiogenesis, DNA replication, and signal transduction are all critical cellular processes, TP53 does not directly influence any of them.
17. **a. Cyclin D/cdk4.** The INK4 family of cdk inhibitors directly inhibits the assembly of cyclin D with cdk4 and cdk6 by blocking the phosphorylation of the cyclin D-cdk4/6 complex. This phosphorylation is necessary for activation of the complex.
18. **e. Phosphorylation of RB1.** Phosphorylation is the attachment of a phosphate to a protein. It alters the conformation of the protein and therefore is an excellent method of regulating gene function. Cyclin-cdk complexes phosphorylate RB1 or its family members, p107 and p130. Phosphorylated RB1 can no longer bind to members of the E2F family of transcription factors. Free E2F heterodimerizes with DP1 or DP2 and transcriptionally activates genes important in DNA replication, such as DNA polymerase-A and the cell cycle, such as E2F-1. Dephosphorylation of RB has the opposite effect. E2F regulation does not occur directly though

phosphorylation or dephosphorylation. MDM2 regulates TP53 and is not directly affected by cyclin-cdk complexes.

19. **c. DNA damage.** Hypoxia, nutrient-poor environment, and cytokines are all signals to the cell that it is not in an environment that is conducive to cellular division. These signals influence the cell BEFORE it invests its energy in replicating the DNA, that is, it is a G1S checkpoint. If DNA damage occurs, it is critical the errors are repaired before cellular division. Therefore, DNA damage can lead to cell cycle arrest at both the G_1S and G_2M checkpoints.
20. **d. Ultraviolet light.** Nucleotide excision repair (NER) is a major defense against DNA damage caused by ultraviolet radiation and chemical exposure. NER acts on a wide range of alterations that result in large local distortions in DNA by recognizing distortions in the DNA helix, excising the damaged DNA, and replacing it with the correct sequence. Base excision repair (BER) is the primary mechanism for repairing damage caused by reactive oxygen species. MMR is the primary mechanism for polymerase errors. Double-strand break repair is the primary mechanism for repairing double-strand breaks.
21. **b. Homologous recombination (HR).** HR is one of two mechanisms for repairing DNA double-strand breaks, the other being non-homologous end joining (NHEJ). In HR, the normal undamaged sister chromatid is used as a template to repair the damaged segment of DNA.
22. **c. BRCA1 (breast cancer type 1 susceptibility protein).** BRCA1 is associated with familial breast and ovarian cancer. It is believed that the breast cancer susceptibility gene BRCA1 (as well as BRCA2) plays an important role in HR as well as sensing DNA damage. Both BRCA1 and BRCA2 are part of an enzymatic complex with RAD51- and BRCA1-associated RING domain 1 (BARD1). This complex is recruited by proliferating cell nuclear antigen (PCNA) to regions that have undergone DNA damage to repair DNA breaks.
23. **e. Proteolytic cleavage.** Procaspases are the larger, inactive precursor forms of the caspase proteins. Specific proteolytic

cleavage is required for their activation. The caspases themselves are proteases and, once activated, proceed to cleave and activate other caspases, thus facilitating a proteolytic cascade that serves to amplify the initial apoptotic signal.

24. **a. Independent of TP53.** The identification of ligand-dependent apoptosis receptors may have a profound impact on therapy. Most cancer therapies (e.g., chemotherapy and external beam radiotherapy) depend on TP53 to induce apoptosis in the cancer cell. Because TP53 is mutated in more than half of malignancies, TP53-independent pathways for apoptosis are of great clinical interest. Because ligand-dependent apoptosis is independent of TP53, these receptors and ligands are attractive and novel treatment targets.
25. **a. APAF-1/caspase 9.** The TP53-induced apoptosis is dependent on the APAF-1/caspase 9 activation pathway.
26. **c. Increasing the mitochondrial membrane permeability.** Although each proapoptotic bcl-2 family member responds to different stimuli, the principal mechanism by which these family members induce cell death is by increasing mitochondrial membrane permeability.
27. **e. Maintaining chromosomal length.** Telomerase immortalizes cells by maintaining the ends of the chromosomes, or telomeres, which normally shorten with each cell division.
28. **c. DNA hypomethylation.** Normal cells closely monitor their telomere lengths and, should they fall below a critical threshold length, will initiate either cell senescence or apoptosis. If key players in these responses (e.g., TP53) are mutated, then chromosomal instability may result contributing to cancer initiation. To date, there is not a strong connection known for telomere loss and loss of DNA methylation.
29. **a. Fusion of BCR to ABL via chromosome translocation.** This particular chromosomal abnormality is typically found in chronic myelogenous leukemia patients, not in solid tumors. TMPRSS2-ERG gene fusions are found in ~50% of prostate cancer cases, MiTF/TFE gene family translocations have been associated with a subset of renal cell carcinomas, isochromosome 8q is found in a

subset of prostate cancers and is often associated with the loss of 8p, loss of chromosome 9 is observed in a subset of urothelial cancers and enumeration of this chromosome by FISH is one of the components of a molecular test used for detecting bladder cancer.

30. **a. Testis.** The urologic malignancy most closely linked to a karyotypic abnormality is a testis tumor. A 12p isochromosome was first identified in a testis tumor in the 1980s, and experimentally this cytogenetic hallmark of testicular tumors has diagnostic and prognostic value.
31. **a. The 8q24 chromosomal region.** Systematic review of GWAS studies in prostate cancer indicate that the 8q24 region continues to be the most implicated in prostate cancer risk, and among different racial cohorts.
32. **e. All of the above.** RNA-seq technologies can unbiasedly sequence all RNA species in a sample, with an analysis that is not limited to "annotated" sequences.
33. **b. Sequencing technologies can identify the presence of organisms that cannot be grown in culture.** This serves as an advantage over standard microbial culture because many microorganisms that are present as members of the human microbiome are not readily culturable with standard clinical microbiology protocols.
34. **d. Nivolumab.** A randomized, open-label, phase 3 study comparing nivolumab with everolimus in patients with renal-cell carcinoma who received previous treatment demonstrated improved overall survival and fewer severe adverse events with nivolumab therapy (Motzer et al., 2015).

Chapter Review

1. Mutations in DNA can lead to changes in protein function or expression that increase the potential for cancer initiation, progression, or metastasis.
2. Tumor suppressor genes normally negatively regulate and control cellular growth. Oncogenes normally promote cell growth.

3. Loss of tumor suppressor gene function can occur primarily by: (1) homozygous deletion, (2) loss of one allele and mutational inactivation of the second allele, (3) mutational events involving both alleles, and (4) loss of one allele and epigenetic inactivation of the second allele.
4. Certain tumor suppressor genes do not follow the "two-hit" hypothesis and may be inactivated via dominant negative mutations or haploinsufficiency.
5. Proto-oncogenes can be converted to oncogenes by: (1) mutation of the proto-oncogene resulting in an activated form of the gene, (2) gene amplification, and (3) chromosomal rearrangement.
6. The cell cycle consists of an ordered, unidirectional series of events, the main goal of which is to replicate the cell's genome and partition one copy into each of two resulting daughter cells.
7. The cell cycle is divided up into 4 phases; G1, S, G2, and M. The transition from G1 into S is critically dependent on phosphorylation of the RB1 tumor suppressor protein. Mutations in *RB1* are common in urologic malignancies.
8. Phase-specific phosphorylation of substrate proteins by cyclin-dependent kinases (cdks) orchestrates progression through the cycle.
9. The activities of cdks are dependent upon their association with specific cyclin proteins. Cyclins accumulate and are rapidly degraded in a phase-specific manner, thus assuring the proper sequencing and irreversibility of key events throughout the cell cycle.
10. Primary points of cell cycle control are the G1S and G2M checkpoints. Checkpoints employ cyclin-dependent kinase inhibitor (CDKI) proteins to pause the cell cycle in response to a variety of signals, including DNA damage, cell-cell contact, cytokine release, and hypoxia.
11. The TP53 tumor suppressor protein is a key player in cell cycle checkpoints, responding to DNA damage by signaling cell cycle arrest and repair of the DNA damage. If the DNA damage cannot be repaired, TP53 may trigger cell death (apoptosis).
12. *TP53* is the most commonly mutated gene in cancer and plays a prominent role in genitourinary malignancies.

13. Defects in cell cycle checkpoints lead to unregulated cell proliferation and genetic instability.
14. Methylation occurs specifically at CpG dinucleotides in the genome. The presence of 5-methylcytosine in DNA can result in spontaneous deamination to thymine and, therefore, the formation of C→T transition mutations.
15. DNA methylation can affect gene function by subsequent mutational events or epigenetic mechanisms. Methylation of CpG islands associated with the promoter region of genes may result in suppression of gene expression.
16. Loss of promoter methylation of normally methylated genes can lead to inappropriate gene expression; for example, expression of oncogenes.
17. DNA damage does not often lead to malignancy because the cell possesses multiple repair mechanisms.
18. Defects in DNA repair facilitate the accumulation of the mutations critical for tumor formation and progression.
19. NER is a major defense against DNA damage caused by ultraviolet radiation and chemical exposure.
20. BER repairs damage caused by spontaneous deamination of bases, radiation, oxidative stress, alkylating agents, and replication errors.
21. MMR removes nucleotides mispaired by DNA polymerase.
22. Double-stranded break repair (DSBR) is a major defense against DNA damage caused by ionizing radiation, free radicals, and chemicals.
23. Many syndromes involving inherited defects in DNA repair exhibit marked increases in cancer susceptibility, strongly linking genomic instability and cancer.
24. Large variations in chromosome numbers and complex structural rearrangements, as well as intra-tumoral variation in these aberrations, are hallmarks of most human solid tumors.
25. The extent of chromosomal abnormalities typically correlates with disease severity and aggressiveness.
26. Recurrent structural rearrangements occur in prostate (ETS gene fusions), renal (MiTF/TFE family translocation carcinomas), and testicular cancers (isochromosome 12p).

27. Copy number alterations in a particular gene, coupled with changes in the other allele, are evidence for that gene functioning as a disease-relevant oncogene or tumor suppressor gene.
28. Genes discovered to have germline mutations that cause familial forms of cancer may also be involved in the sporadic form of the disease (e.g., VHL in ccRCC).
29. High-density single nucleotide polymorphism (SNP) microarrays have been used in genome-wide association studies (GWAS) to identify DNA sequence variants associated with cancer risk.
30. Low-grade bladder cancers that progress are typically high-grade papillary tumors of the luminal molecular subtype, and high-grade bladder cancers that originate de novo are more consistent with basal-like expression.
31. Telomeres contain stretches of terminal, non-coding, repetitive DNA that cap the ends of each chromosome, thereby stabilizing them.
32. Telomere DNA repeats are progressively lost as cells divide and as a result of oxidative DNA damage at the telomeres.
33. Normal cells monitor their telomere lengths and permanently exit the cell cycle (cellular senescence) or commit suicide (apoptosis) in tumor-suppressive responses to telomere shortening. This telomere length checkpoint involves TP53 and RB1.
34. Loss of telomere length checkpoints leads to critical telomere shortening, which initiates chromosomal instability contributing to cancer initiation.
35. Most cancers and pre-malignant lesions have abnormally short telomeres.
36. Most cancers express the enzyme telomerase, which re-stabilizes the telomeres and allows unlimited cell division potential (immortalization); thus telomerase represents an attractive therapeutic target.
37. Apoptosis is a rapid, orderly programmed form of cell death that is utilized by multicellular organisms to eliminate unwanted cells.
38. Apoptosis is believed to play an important role in tumor suppression, as many of the signals that induce apoptosis arise from potentially tumorigenic cell stresses such as DNA damage.

39. Cancer is characterized by interruptions in the normal process of apoptosis, resulting in inappropriate cell survival.
40. Apoptosis is mediated by a conserved family of proteases known as caspases. Initiator caspases start caspase proteolytic cascades terminating in the activation of executioner caspases that target several cellular proteins.
41. Two main apoptotic pathways have been identified. In the intrinsic pathway, Bcl-2 family members modulate the release of cytochrome c from the mitochondria, which participates in the activation of initiator caspases. The extrinsic pathway activates caspases in response to signals from extracellular "death receptors."
42. In addition to its functions in cell cycle arrest and DNA repair, TP53 plays a key role in apoptosis.
43. Bcl-2 is a classic inhibitor of the mitochondrial pathway of apoptosis and is overexpressed in some genitourinary malignancies.
44. Therapeutic response is often dependent upon the integrity of apoptotic pathways in the cancer cells. Most testicular germ cell tumors (TGCT) retain intact DNA damage response (DDR), wild type TP53, and apoptotic responses, providing high cure rates with DNA damaging agents.
45. Novel agonists and antagonists of apoptosis, such as ceramide and clusterin, may successfully be controlled to combat cancer.
46. Stem cells are defined by their ability to differentiate along multiple lineages and their immortality.
47. Cancer is believed to be a stem cell disease in which a small population of cancer stem cells maintains the larger tumor.
48. Cancer may ultimately be eradicated by targeting only the cancer stem cell.
49. The PD-1 pathway limits immune activity resulting in decreased autoimmunity and cytokine.
50. Inhibition of the interaction between PD-1 and PD-L1 can enhance T-cell response and mediate antitumor activity.
51. PD-1/PD-L1 blockade for treatment of advanced or metastatic bladder cancer has demonstrated efficacy as first-line treatment for cisplatin-ineligible patients.

52. Nivolumab has been shown to improve overall survival with fewer adverse events compared to everolimus in patients with RCC who received previous treatment.
53. Loss-of-function mutations in PBRM1 were associated with clinical benefit among patients with metastatic clear cell RCC who received anti-PD-1 therapy.
54. The clinical dogma that urine is sterile has been challenged by studies describing distinct microbiota present in urine that are representative of species residing within the urinary tract. These microorganisms would have previously been under or unrecognized using standard clinical microbiology culture techniques.
55. Direct interactions between the urinary microbiome and genitourinary cancer development or progression is now an area of active study.
56. Indirect interactions between the microbiome, such as the GI microbiota, and genitourinary cancers is also an area of active research. The GI microbiota may have a strong influence on treatment response and/or related toxicities to cancer therapies.